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(54) **SOLENOID VALVE**

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*Primary Examiner* — Craig J Price

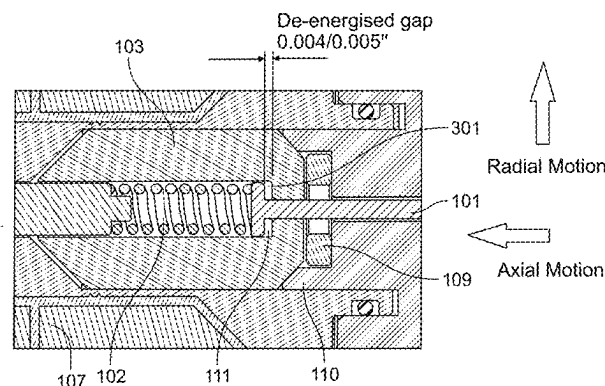
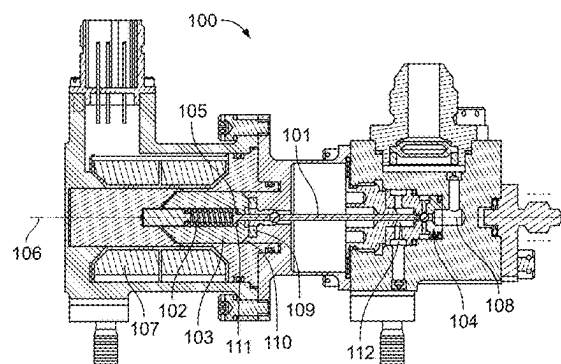
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**ABSTRACT**

A solenoid valve for controlling a fluid flow comprises an actuating member moveable between a first position and a second position. Movement of the actuating member in one direction effects opening of the valve and movement of the actuating member in an opposite direction effects closing of the valve. The solenoid valve comprises a biasing member for biasing the actuating member towards the first position; and an armature moveable in response to energising/de-energising of the solenoid valve. Energising the solenoid valve is operable to move the armature to engage the actuating member and cause movement of the actuating member to the second position against the biasing member. De-energising the solenoid valve allows the biasing member to effect movement of the armature and the actuating member towards the first position, initially causing disengagement of the armature from the actuating member, and, subsequently causing retainment of the armature in a resting position.

**13 Claims, 3 Drawing Sheets**



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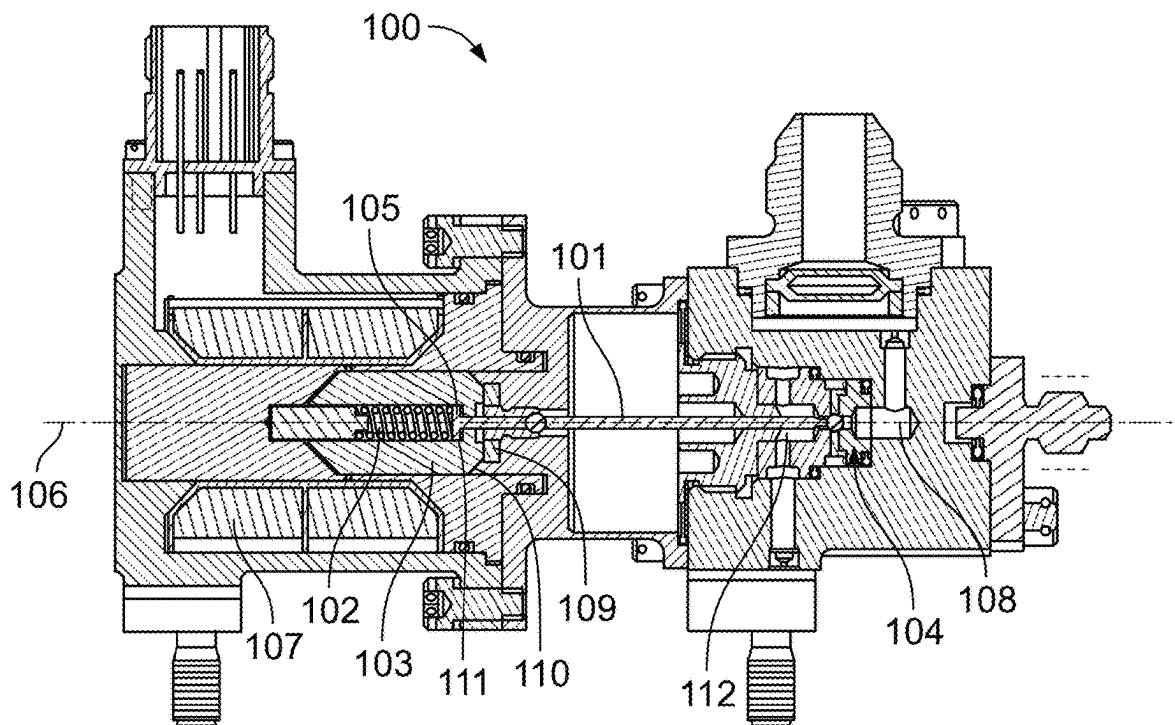


Figure 1

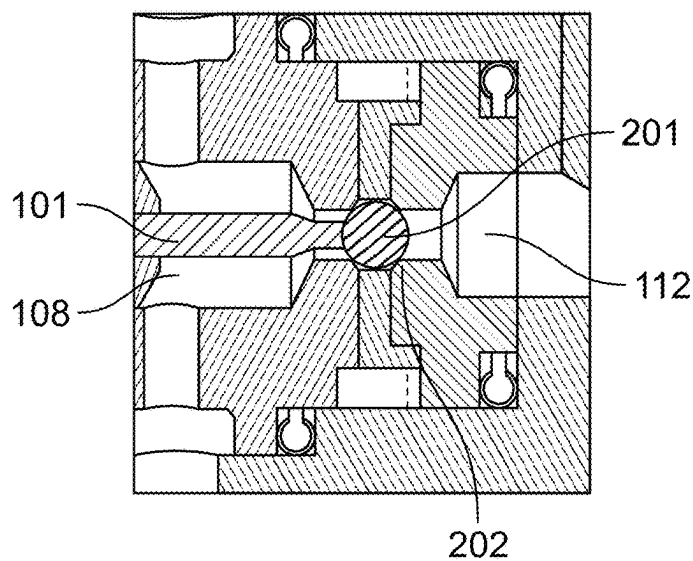


Figure 2

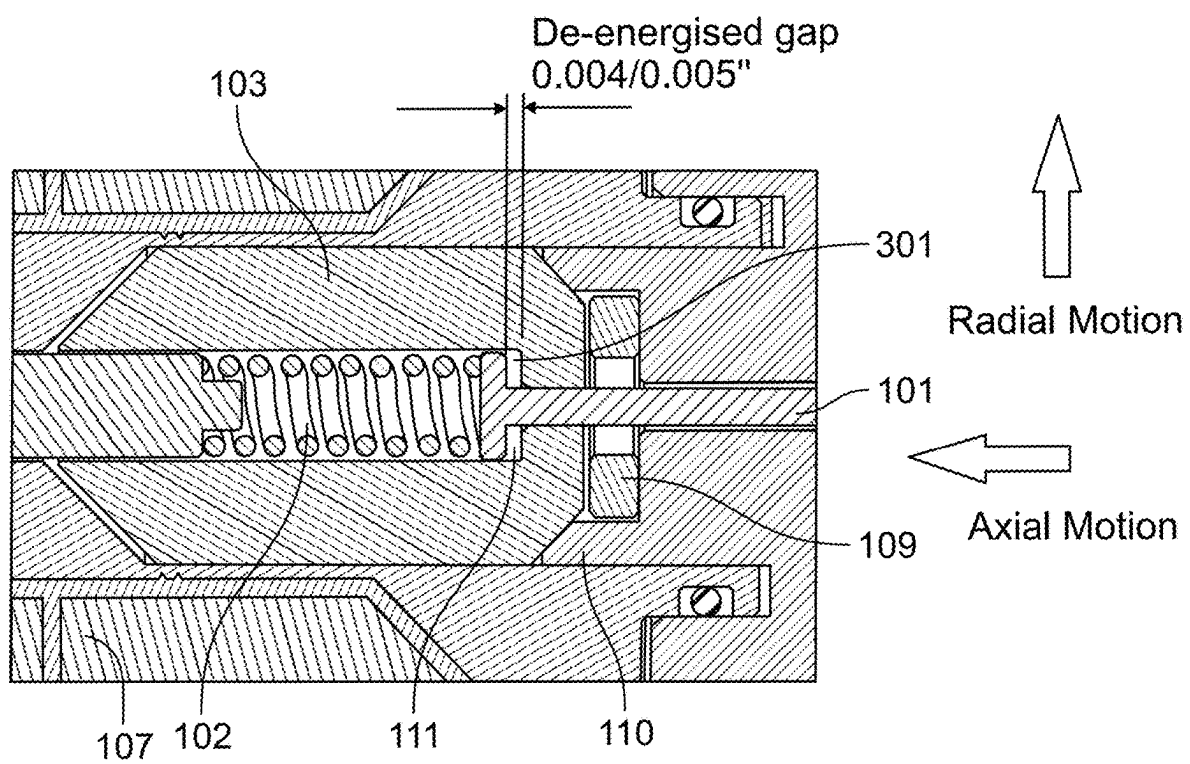


Figure 3

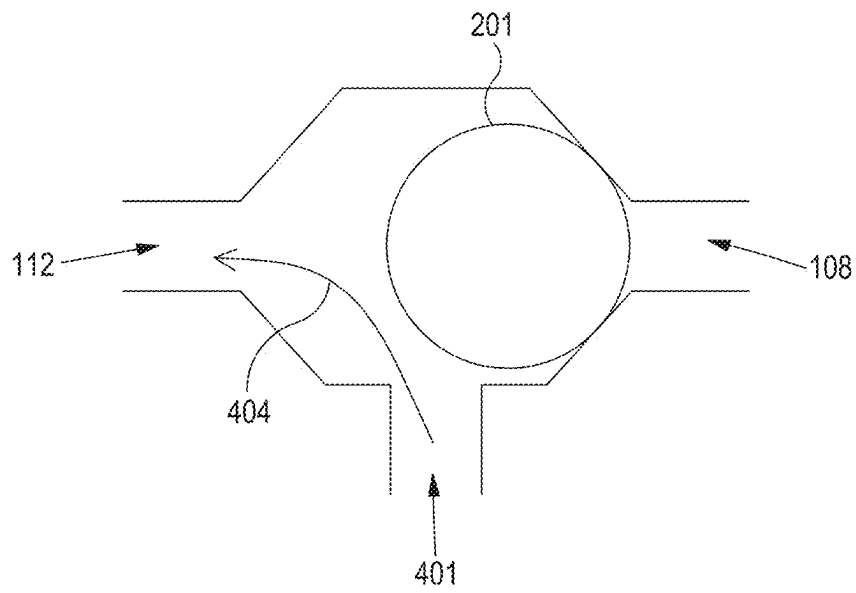


Figure 4

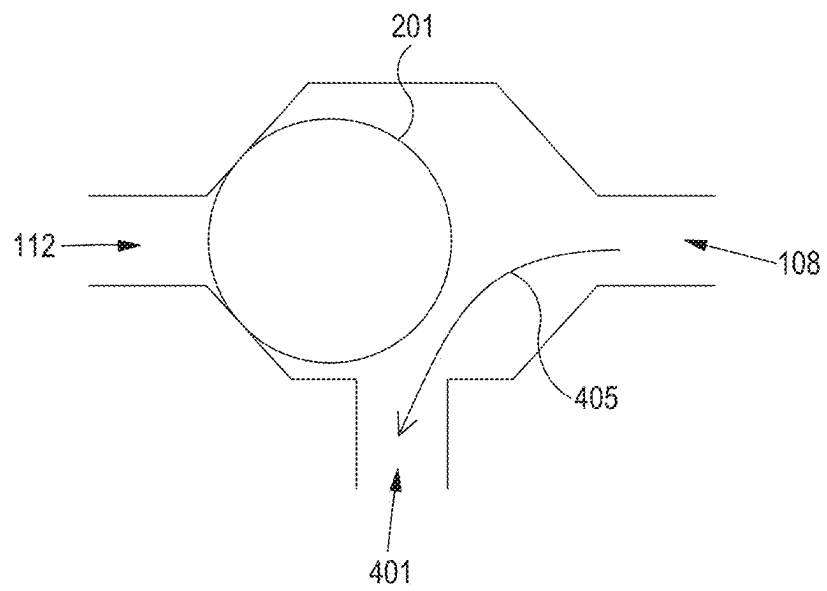


Figure 5

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**SOLENOID VALVE****CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is a U.S. national stage application, filed under 35 U.S.C. § 371, of International Application No. PCT/EP2020/060274, filed on Apr. 9, 2020, which claims priority to United Kingdom Application No. 1905129.1 filed Apr. 11, 2019; the entire disclosures of which are incorporated herein by reference in their entireties for all purposes.

**FIELD OF INVENTION**

The invention relates to solenoid valves, in particular, solenoid valves used for controlling a secondary air flow in a jet engine.

**INTRODUCTION**

Solenoid valves are used to control a flow of fluid. Typically, when solenoid valves are energised or de-energised, flow is allowed or prevented. Solenoid valves usually include an actuator that is moved by creating an electromagnetic field. Movement of the actuator controls opening and closing of the valve.

Solenoid valves are used in extreme environments such as to rapidly control a secondary air flow to a secondary air system of a jet engine. Here, solenoid valves are subject to vibration, extreme acceleration, and extreme temperature environments. For example, such solenoid valves may be subject to temperatures of around 600 degrees centigrade. These extreme conditions can cause solenoid valves to not function properly. For example, material properties can degrade at high temperatures meaning that forces caused by contact between moving parts can cause increased wear and damage than would otherwise be the case in a less extreme environment.

In addition, extreme changes in temperature undesirably affects stroke clearances of sub components due to differential thermal expansion.

It is an object of the present invention to address these problems and to provide an improved solenoid valve.

**SUMMARY**

According to a first aspect of the invention there is provided a solenoid valve for controlling a fluid flow. The solenoid valve comprises: an actuating member moveable between a first position and a second position, wherein movement of the actuating member in one direction effects opening of the valve and movement of the actuating member in an opposite direction effects closing of the valve; a biasing means such as a spring for biasing the actuating member towards the first position; and an armature moveable in response to energising/de-energising of the solenoid valve. Energising the solenoid valve is operable to move the armature to engage the actuating member and cause movement of the actuating member to the second position against action of the biasing means. De-energising the solenoid valve allows the biasing means to effect movement of the armature and of the actuating member towards the first position, initially causing disengagement of the armature from the actuating member, and, subsequently causing retainment of the armature in a resting position. Preferably,

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this retainment is magnetic retainment but other methods of retainment may be used such as a disengageable hook connection.

The biasing means may be a biasing member or an elastic object/substance that stores mechanical energy.

Advantageously, the actuating member is disengaged (or de-coupled) from the armature during movement of the actuating member towards the first position when the valve is de-energised. This means that the armature and actuating member are not in contact when the armature is stopped at the resting position. Any kinetic energy removed from the armature when the armature is stopped is advantageously not transferred to the actuating member.

Opening or closing (but preferably closing) of the valve is effected by the force of the biasing means after the actuating member is disengaged from the armature. The actuating member and any valve opening/closing mechanisms that are connected to the actuating member are therefore subject to less, or even, no impact stress caused by the stopping of the armature at the resting position.

The present invention reduces the amount of energy transferred to the actuating member when the valve is to be opened or (preferably) closed as a result of the solenoid valve being de-energised. However, when the solenoid valve is energised, there is still sufficient energy transferred to the armature to overcome any forces maintaining the armature in the resting position, and, to cause movement of the actuating member against the biasing means after the actuating member is engaged by the armature.

The reduction in energy transferred to the actuating member and resulting reduction of impact force to the actuating member means that degradation of components or materials in the solenoid valve is reduced. This effect is particularly important in mitigating degradation to the actuating member and any corresponding valve mechanism caused by impact forces and exacerbated by the extreme environment such as high temperature.

Optionally, the solenoid valve further comprises a permanent magnet providing a magnetic force sufficient to magnetically retain the armature in the resting position.

Preferably, energising the solenoid valve is operable to provide sufficient electromagnetic force to the armature to overcome the magnetic force in order for the armature to move away from the resting position.

Advantageously, magnetic retainment of the armature in the resting position, such as by using the optional permanent magnet, reduces or even prevents any subsequent substantial movement of the armature when in the resting position and the solenoid valve is de-energised. Without any such retainment, the armature would be free to move under acceleration loads when the valve is de-energised because the armature is decoupled from the actuation member. This could cause allow significant vibration and/or oscillation of the armature of a valve mounted in, for example, a jet engine. Such vibration could cause excessive wear and failure and is prevented using magnetic retainment such as the permanent magnet. The armature itself may be at least partially magnetic (or ferromagnetic) and therefore attracted to the permanent magnet.

Preferably, the magnetic force provided by the permanent magnet is sufficient to retain the armature in the resting position when the solenoid valve is under acceleration loads (such as those experienced by components in a jet engine of an aircraft). The armature may be retained to such a degree that it is not subject to vibration under acceleration loads.

The actuating member may be configured to cause closure of the valve when the actuating member is in the first

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position and to cause opening of the valve when the actuating member moves towards the second position.

Optionally, the actuating member and the armature are both configured to move along a common axis.

Optionally, the contact surface of the armature is spaced from the actuating member by a de-energised gap when the armature is in the resting position and the actuating member is in the first position.

Optionally, the de-energised gap is between 0.002 inches and 0.010 inches (0.051 mm to 0.254 mm). Preferably, the de-energised gap is between 0.004 inches and 0.005 inches (0.100 and 0.130 mm).

Optionally, the actuating member comprises a stem extending into an internal space within the armature. Optionally, the contact surface of the armature is within the internal space of the armature and is configured to abut with a head portion of the stem when the armature engages the actuating member.

This arrangement enables the armature to engage with the actuating member whilst being surrounded by solenoid coils for providing the electromagnetic field that causes movement of the armature whilst providing for a compact design of the solenoid valve.

Optionally, the solenoid valve further comprises a cavity surrounded by solenoid coils, and, wherein the armature is configured to axially move within the cavity due to the electromagnetic force generated when a voltage is applied to the solenoid coils.

Optionally, the biasing means is a spring attached between a fixture point and the actuating member. The spring may extend through the internal space of the armature between the head portion and a fixture point in the cavity.

Optionally, the solenoid valve further comprises a stopping member comprising a frusto-conical surface configured to axially stop an inner portion of the armature when the armature reaches the resting position. Optionally, the inner portion of the armature is shaped to substantially conform to the frusto-conical surface.

Whilst the permanent magnet discussed above reduces axial motion, the cone formed by the frusto-conical surface helps to reduce or even prevent radial motion and rocking of the armature, thereby further reducing wear on the armature.

Optionally, the actuating member is configured to maintain a valve ball on a valve seat when the actuating member is in the first position thereby preventing a fluid flow out of an air inlet chamber, and, when the actuating member is in the second position, the actuating member is configured to allow the valve ball to be released from the valve seat by pressure of a fluid thereby allowing a the fluid to flow out of an air inlet chamber.

The features discussed above are particularly advantageous for use with a valve ball and seat. A valve ball and seat arrangement is particularly susceptible to wear caused by impact stress on the valve ball or valve seat that could be caused by dissipation of kinetic energy from the armature if the armature was not decoupled from the actuation member. The decoupling of the armature and actuation member reduces impact stresses on the valve ball and valve seat thereby rendering the solenoid valve more suitable for use in high temperature environments such as in a jet engine.

Optionally, the solenoid valve is configured to allow or prevent a secondary air flow to a secondary air system in a jet engine.

There is also disclosed a pneumatic solenoid valve as described above wherein the actuating member is movable to open the valve in response to a fluid pressure acting against the actuating member and biasing means.

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According to a second aspect of the invention there is provided a method of controlling a fluid flow comprising the steps of:

moving an actuating member between a first position and a second position, wherein movement of the actuating member in one direction effects opening of the valve and movement of the actuating member in an opposite direction effects closing of the valve;

providing a biasing means for biasing the actuating member towards the first position;

moving an armature in response to energising/de-energising of the solenoid valve;

energising the solenoid valve to move the armature to engage the actuating member and cause movement of the actuating member to the second position against action of the biasing means; and

de-energising the solenoid valve to allow the biasing means to effect movement of the armature and of the actuating member towards the first position, initially causing disengagement of the armature from the actuating member, and, subsequently causing retainment of the actuating member in a resting position.

#### BRIEF DESCRIPTION OF THE FIGURES

FIG. 1 is a cross-sectional representation of a solenoid valve when in a de-energised state according to this disclosure.

FIG. 2 is a cross-sectional magnified view of a valve mechanism of the solenoid valve of FIG. 1.

FIG. 3 is a cross-sectional magnified view of an armature of the solenoid valve of FIG. 1.

FIG. 4 is a schematic view of an example valve mechanism of the solenoid valve of FIG. 1 when in the de-energised state (valve closed).

FIG. 5 is a schematic view of an example valve mechanism of the solenoid valve of FIG. 1 when in the energised state (valve open).

#### DETAILED DESCRIPTION

With reference to FIGS. 1 to 3 there is disclosed a solenoid valve **100**. Components of the solenoid valve **100** are movable along axis **106** in either direction. This is referred to as axial movement.

An actuating member **101** is connected between a spring **102** and a valve mechanism **104**. In this example, the valve mechanism **104** is a valve ball **201** and valve ball seat **202** (as shown in FIG. 2) but other arrangements could be utilised. A head portion **105** of the actuating member **101** is located within an internal cavity of an armature **103**, the armature **103** usually comprising a ferromagnetic material. The head portion **105** is flanged so that it abuts with an internal surface **111** within the internal cavity of armature **103** when the head portion **105** and armature **103** are at certain axial positions relative to one another. The actuating member **101** is biased towards a first position (which is the position as shown in the Figures) and is axially movable to a second position (not shown) against the action of the spring **102** which, in the shown example, effects opening of the valve.

The armature **103** is movable axially within the cavity of the solenoid valve under electromagnetic force provided by coils **107**. For example, when an electric current flows through coils **107** (i.e. when the solenoid is energised), the armature **103** is pulled against the biasing force of the spring **102**. The armature **103** is shown in a resting position in the

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Figures and is movable in response to energising/de-energising of the solenoid. It can be seen from the Figures that movement of the armature **103** in an axial direction (to the left in the example of FIG. 1) would cause contact of the internal surface **111** of the armature **103** with the head portion **105** of the actuating member **101**.

A permanent magnet **109** is located axially adjacent to the armature **103** when the armature **103** is in the resting position. In the shown example, the armature **103** radially surrounds the actuating member **101**. The permanent magnet **109** is configured to magnetically retain the armature **103** in a fixed position. However the armature **103** can still move in response to energising of the solenoid since the energising provides sufficient electromagnetic force for the armature **103** to overcome the magnetic attraction provided by the permanent magnet **109**.

A stopping member having a frusto-conical surface **110** is also located axially adjacent to the armature **103**. In the shown example, the frusto-conical surface **110** radially surrounds the permanent magnet when the armature **103** is in the resting position. The frusto-conical surface **110** conforms to a corresponding surface of the armature **103** such that radial movement of the armature **103**, for example due to vibration, is minimised.

In operation, the solenoid valve controls a fluid flow, for example enabling air to flow i) in a first direction from a common source to a vent **112** or ii) in a second direction from a common source to an air inlet chamber **108** as will be described further below. The direction of air flow depends on whether the solenoid valve is in an energised state or a de-energised state. These states will be described below in more detail:

The solenoid valve as shown in FIGS. 1 to 3 is in a de-energised state. In this condition the actuating member **101** is in a first position, in which in this example the valve is closed. In particular, with reference to FIG. 3, there is a de-energised gap **301** in between the head of the actuating member **101** and the internal surface **111** of the actuator. The armature is therefore de-coupled from the actuating member **101** that operates the valve mechanism **104**. The axial distance of the de-energised gap may be between 0.004 and 0.005 inches (0.100 and 0.130 mm). The axial distance is sufficient to minimise any further contact between the armature **103** and actuating member **101** after the decoupling (for example caused by vibration). In the de-energised state, the armature **103** is held in position by the magnetic force of the permanent magnet **109** and the abutment of the armature **103** with the cone **110**. These features restrict movement of the armature. Therefore, the armature **103** is prevented from being subject to undesirable vibration or motion caused by acceleration loads on the solenoid valve that, for example, components in a jet engine would typically be subjected to. Such undesirable vibration or motion causes excessive wear that would be exacerbated by the extremely high temperatures that such components are subjected to, for example, around 600 degrees centigrade.

When the solenoid valve is energised to enter the energised state, the armature **103** is configured to axially move due to the electromagnetic force generated by the solenoid coils **107**. The armature **103** moves towards the head **105** of the actuating member **101** thereby closing the de-energised gap. The solenoid coils **107** provide sufficient electromagnetic force to overcome the magnetic force of the permanent magnet **109** that is holding the armature in the resting position. Once the armature **103** contacts the head **105** of the actuating member **101**, the actuating member **101** is drawn away from the first position against the force of the spring

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**105**, thereby compressing the spring **105** and building potential energy. The actuating member **101** is now in the second position. At this point, the valve mechanism **104** is allowed to open due to the pressure of air in the air inlet chamber **108**. In some embodiments, the solenoid valve may be configured to enable the valve to be opened merely due to an extremely high pressure in the inlet chamber **108** that is sufficient to force the actuation member against the spring **105**, for example for preventing pressure in the air inlet chamber **108** from exceeding an acceptable level whether or not the solenoid valve is energised.

When the solenoid valve is de-energised, the potential energy that has built up in the (now compressed) spring **105** is released thereby forcing both the armature and the actuating member **101** back towards the first position. The actuating member **101** causes closure of the valve mechanism **104**. The closure of the valve mechanism **104** is accomplished due to the biasing force provided by the spring **105**. Meanwhile, the armature **103** continues moving due to one or both of momentum acquired from the spring **105** and the magnetic attraction of the permanent magnet **109**. Eventually, the armature **103** will be stopped by coming into contact with the frusto-conical surface **110** of the stopping member. The kinetic energy of the armature **103** is advantageously dissipated on the abutment member because the armature **103** has become decoupled from the actuating member **101**. Therefore, kinetic energy of the armature **103** is not dissipated on the valve mechanism **104** which would lead to excessive wear of the valve mechanism **104** and even damage.

A possible arrangement for the valve mechanism **104** is now described in more detail with reference to FIGS. 4 and 5. The valve mechanism includes a common source **401** for providing high pressure fluid flow. Depending on the position of the ball **201**, the fluid can be directed to flow from either i) the common source **401** to a vent **112** as shown in FIG. 4 and indicated by arrow **404**, or ii) the inlet chamber **108** to the common source **401** as shown in FIG. 5 and indicated by arrow **405**. When the actuating member **101** is in the first position, the ball **201** is held in the position shown in FIG. 4 by the force of the biasing means **102** enabling fluid to flow in the direction **404** thus preventing fluid flow out of the inlet chamber **108**. When the actuating member **101** is in the second position, the ball **201** is pushed to the position shown in FIG. 5 by pressure provided by the inlet **108** and/or the common source **401** thus enabling fluid to flow out of the inlet **108** to the common source **401**.

The invention improves valve component life and impact stress on the ball, valve seat, and other components transferring load from the spring **102** is reduced. The solenoid valve is therefore more suitable for use in high temperature environments, since material properties degrade at elevated temperature and are more susceptible to wear/damage caused by undesirable impact stress.

In the example discussed above, the first position of the armature **103** corresponds to closing the valve **104** and the second position of the armature **103** corresponds to opening the valve **104**. However in other examples the first position of the armature **103** corresponds to opening the valve **104** and the second position of the armature **103** corresponds to closing the valve **104**. For example, the valve mechanism may be a ball and seat arrangement located on the left side of the armature **103** whilst the spring **102** and head **105** of the actuating member have the same arrangement shown in the Figures. In this case, energising the solenoid valve would cause compression of the spring and closure of the valve.



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De-energising the solenoid valve would cause opening of the valve due to the biasing force of the spring.

The invention claimed is:

1. A solenoid valve for controlling a fluid flow, the solenoid valve comprising:

an actuating member moveable between a first position and a second position, wherein movement of the actuating member in one direction effects opening of the solenoid valve and movement of the actuating member in an opposite direction effects closing of the solenoid valve;

a biasing means having an equilibrium position configured to bias the actuating member towards the first position; and

an armature moveable in response to energising/de-energising of the solenoid valve, wherein;

energising the solenoid valve is operable to move the armature to engage the actuating member and cause movement of the actuating member to the second position against action of the biasing means; and

de-energising the solenoid valve causes the biasing means to return to the equilibrium position thereof and to effect movement of the armature and of the actuating member towards the first position, initially causing disengagement of the armature from the actuating member, and, subsequently causing retainment of the armature in a resting position, and wherein the solenoid valve further comprises a permanent magnet located axially adjacent to the armature when the armature is in the resting position and providing a magnetic force sufficient to magnetically retain the armature in the resting position, and, wherein energising the solenoid valve is operable to provide sufficient electromagnetic force to the armature to overcome the magnetic force in order for the armature to move away from the permanent magnet.

2. The solenoid valve of claim 1 wherein the magnetic force provided by the permanent magnet is sufficient to retain the armature in the resting position when the solenoid valve is under acceleration loads.

3. The solenoid valve of claim 1 wherein the actuating member is configured to cause closure of the solenoid valve when the actuating member is in the first position, and, wherein the actuating member is configured to cause opening of the solenoid valve when the actuating member moves towards the second position.

4. The solenoid valve of claim 1 wherein the actuating member and the armature are both configured to move along a common axis.

5. The solenoid valve of claim 1 wherein a contact surface of the armature is spaced from the actuating member by a de-energised gap when the armature is in the resting position and the actuating member is in the first position.

6. The solenoid valve of claim 5 wherein the de-energised gap is between 0.002 inches and 0.010 inches (0.051 mm to 0.254 mm).

7. The solenoid valve of claim 6, wherein the de-energised gap is between 0.004 inches and 0.005 inches (0.100 and 0.130 mm).

8. The solenoid valve of claim 1 wherein the actuating member comprises a stem extending into an internal space

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within the armature, and, wherein a contact surface of the armature is within the internal space of the armature and is configured to abut with a head portion of the stem when the armature engages the actuating member.

9. The solenoid valve according to claim 1 further comprising a cavity surrounded by solenoid coils, and, wherein the armature is configured to move within the cavity due to the electromagnetic force generated when a voltage is applied to the solenoid coils.

10. The solenoid valve according to claim 1 wherein the biasing means is a spring.

11. The solenoid valve according to claim 1 further comprising a stopping member having a frusto-conical surface configured to axially stop an inner portion of the armature when the armature reaches the resting position, and wherein the inner portion of the armature is shaped to substantially conform to the frusto-conical surface.

12. The solenoid valve according to claim 1 wherein, the actuating member is configured to maintain a valve ball on a valve seat when the actuating member is in the first position thereby preventing a fluid flow out of an air inlet chamber, and, when the actuating member is in the second position, the actuating member is configured to allow the valve ball to be released from the valve seat by pressure of a fluid thereby allowing the fluid to flow out of an air inlet chamber.

13. A method of controlling a fluid flow comprising the steps of:

moving an actuating member between a first position and a second position, wherein movement of the actuating member in one direction effects opening of a solenoid valve and movement of the actuating member in an opposite direction effects closing of the solenoid valve;

providing a biasing means having an equilibrium position configured to bias the actuating member towards the first position;

moving an armature in response to energising/de-energising of the solenoid valve;

energising the solenoid valve to provide sufficient electromagnetic force to the armature to overcome a magnetic force between the armature and a permanent magnet in order for the armature to move away from the permanent magnet and to engage the actuating member and cause movement of the actuating member to the second position against action of the biasing means; and

de-energising the solenoid valve to cause the biasing means to return to the equilibrium position thereof and to effect movement of the armature and of the actuating member towards the first position, initially causing disengagement of the armature from the actuating member, and, subsequently causing retainment of the actuating member axially adjacent to the permanent magnet in a resting position, the permanent magnet providing the magnetic force sufficient to magnetically retain the armature in the resting position.

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